



Probability and Statistics

Assignment 5

Third Semester

Academic year 2025-2026

Name	Academic number	Section	Serial Number

DIRECTIONS:

- **Deadline for delivery: *Wednesday 24/8/2026.***
- **The delivery date must be adhered to and no duty is accepted after that date.**

45. Dr. Susan Benner is an industrial psychologist. She is currently studying stress among executives of Internet companies. She has developed a questionnaire that she believes measures stress. A score above 80 indicates stress at a dangerous level. A random sample of 15 executives revealed the following stress level scores. 

94	78	83	90	78	99	97	90	97	90	93	94	100	75	84
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- Find the mean stress level for this sample. What is the point estimate of the population mean?
 - Construct a 95 percent confidence level for the population mean.
 - Is it reasonable to conclude that Internet executives have a mean stress level in the dangerous level, according to Dr. Benner's test?
46. As a condition of employment, Fashion Industries applicants must pass a drug test. Of the last 220 applicants, 14 failed the test. Develop a 99 percent confidence interval for the proportion of applicants that fail the test. Would it be reasonable to conclude that more than 10 percent of the applicants are now failing the test?
13. The mean income per person in the United States is \$40,000, and the distribution of incomes follows a normal distribution. A random sample of 10 residents of Wilmington, Delaware, had a mean of \$50,000 with a standard deviation of \$10,000. At the .05 level of significance, is that enough evidence to conclude that residents of Wilmington, Delaware, have more income than the national average?